



**Ahmedabad
University**

CSE 623 Machine Learning

Weekly Report - 3

Group - Vector Minds

Team

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Project Title: Crop Yield Prediction Using Classical Machine Learning Models and Climatic Factors

Project Definition

Agriculture plays an important role in our lives, especially in a well-known agricultural country like India. It provides food security to a larger population; however, its productivity depends heavily on factors such as rainfall and seasonal weather conditions. Due to increasing climate variability, predicting crop yields often becomes difficult.

The aim of this project is to develop a predictive system that estimates crop yield based on provided agricultural and historical data. The project applies classical machine learning regression models to analyze the effect of environmental factors on agricultural productivity.

The project treats crop yield prediction as a supervised regression problem.

Where:

Input Variables (Features): State, Crop type, Season, Crop Year, Area of cultivation, Annual Rainfall, Fertilizer usage, and Pesticide usage.

Output Variable (Target): Crop Yield (production per unit area).

Objectives for Week 3

The main objective of Week 3 of the project was to implement four regression models for crop yield prediction and evaluation of performance - Decision Tree Regressor, Random Forest Regressor, Gradient Boosting Regressor, and Support Vector Regressor (SVR). We put our focus on building a complete machine learning workflow including data preprocessing, model training and evaluation and also apply a 5-fold cross-validation (to assess model robustness and generalization capability), and result visualization. Also we aimed to perform a feature importance analysis using random forest and then generate residual plots and actual v/s predicted plots for all four models that we tested and compare them using MAE, RMSE and R^2 scores. This was to identify which model is the best fit for predicting crop yield using different features.

Work Completed

This week, we made much progress in implementation and evaluation of different machine learning models for crop yield prediction. The dataset was first loaded and cleaned and preprocessing steps were carried out to improve data quality by removing duplicates and handling missing values. Further, the dataset was divided into input features and target variables. Here, the input features for the machine learning models were the selected features and crop yield acted as the output variable.

After preprocessing, using a 80-20 train-test split, the dataset was split where the training set was used for model learning, while the testing set was reserved for evaluating the predictive performance of the models on unseen data. The models used in this week are Decision Tree Regressor, Random Forest Regressor, Gradient Boosting Regressor, and Support Vector

Regression (SVR). These were all implemented using Scikit-learn's pipeline which made the training and preprocessing to occur in a single streamlined workflow.

The **Decision Tree Regressor** was implemented as a basic tree-based model which is capable of capturing non-linear relationships in the dataset. Parameters such as `max_depth=10` and `min_sample_split=10` were required for splitting as they helped in controlling overfitting whilst also preserving the ability to learn non-linear patterns. A shallow version with `depth=3` was also trained separately for visual interpretability.

The **Random Forest Regressor** was implemented as an ensemble of 100 decision trees which was trained using bootstrap sampling with `max_depth=15` and `min_samples_split=10`. This model improves prediction accuracy by reducing variance and increasing robustness and each tree is built on a random subset of features which reduces the variance through averaging of predictions.

The **Gradient Boosting Regressor** was also implemented as one another ensemble learning approach where unlike Random Forest, an ensemble of 100 trees was trained sequentially with a `learning_rate=0.1` and `max_depth=5` with each new model attempting to correct the prediction errors made by the previous model. This iterative learning process helps improve prediction accuracy for complex datasets.

Support Vector Regression (SVR) was implemented using a radial basis function (RBF) kernel where `C=100`, `gamma=0.1` and `epsilon=0.1`, this attempts to find an optimal hyperplane that best fits the data whilst also minimizing prediction error within a defined margin.

Once the models were trained, their performance was evaluated using some metrics like Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score.

5-fold cross-validation was performed to assess the reliability of the tree-based models (Decision Tree, Random Forest, and Gradient Boosting). SVR was not tested because of its high computational costs. Cross-validation divides the dataset into multiple subsets and evaluates the model across different training and validation splits. This process helps in ensuring that the models generalize well to unseen data and do not undergo overfitting, also mean R^2 and standard deviation were recorded for each model.

Feature importance analysis was done next using the random forest model where feature importances were extracted to identify variables with the greatest influence on crop yield prediction. The OneHotEncoder output feature names were also retrieved to label all encoded columns alongside numerical features. A visualization was created with the top 20 important features affecting the model's predictions.

Visualizations were also generated to analyze the behavior and performance of the implemented models. **Actual vs Predicted plots (2x2)** were created to compare predicted crop yields with the actual values in the test dataset where points closer to the diagonal reference line indicated better accuracy. In addition **residual plots (2x2)** were generated to check random distribution of prediction errors around zero and also check for patterns that might indicate model bias.

Model comparison bar charts (MAE, RMSE, R^2) were also generated to provide a clear visual comparison across all four models.

Decision Tree visualization was generated at depth=3 to better understand prediction making by tree-based models. A shallow decision tree was plotted to illustrate the hierarchical decision structure used by the model when predicting crop yield.

Key Observations

The experiments conducted during this week showed that ensemble tree-based models, particularly Random Forest and Gradient Boosting, showed significantly higher R^2 scores and lower MAE and RMSE values compared to the linear models when compared to models used in week 2. These models were able to capture complex relationships between climatic variables and crop yield more effectively. The Decision Tree achieved moderate performance but had more clear interpretability which is useful for prediction explanation. SVR produced competitive results but required careful feature scaling and also was considerably slower to train on the full dataset compared to the tree-based models. Feature importance analysis also indicated that variables such as rainfall, fertilizer usage, and cultivated area play an important role in determining crop productivity. Cross-validation results showed that Random Forest and Gradient Boosting maintained stable R^2 scores across all five folds, indicating robust generalization with low variance across different data splits. Residual analysis revealed that all four models had residuals approximately centered around zero, though some heteroscedasticity was observed at higher yield values, suggesting that high-yield crops may require further specialized modeling.

Tasks Planned for Week 4

The next phase of the project will focus on improving model performance and conducting deeper analysis. Planned tasks include performing hyperparameter tuning using techniques such as GridSearchCV and RandomizedSearchCV, implementing advanced ensemble methods such as stacking models, and conducting further feature importance analysis. Additional visualizations and evaluation techniques will also be used to better interpret model performance and prediction

behavior. Construction of a Stacking Ensemble model combining Random Forest, Gradient Boosting, and Ridge Regression as base learners, with Ridge Regression as the meta-learner. Learning curve analysis to diagnose whether the best model suffers from overfitting or underfitting and to assess whether additional training data would be beneficial. Prediction error distribution analysis to evaluate the symmetry and spread of errors for the best-performing model.

The goal for the upcoming week is to develop a more optimized machine learning model and identify the best performing approach for crop yield prediction.